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Вариант 1

Номер задачи №1 - 1. (Для набора данных проведите кодирование одного (произвольного) категориального признака с использованием метода "count (frequency) encoding")

Номер задачи №2 - 21. (Для набора данных проведите масштабирование данных для одного (произвольного) числового признака с использованием масштабирования по медиане.)

Доп. задание - для произвольной колонки данных построить график "Скрипичная диаграмма (violin plot)"

Задача 1

Номер задачи №1 - 1. (Для набора данных проведите кодирование одного (произвольного) категориального признака с использованием метода "count (frequency) encoding")

In [ ]:
```
!pip install category_encoders
from category_encoders.count import CountEncoder as ce_CountEncoder
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import RobustScaler
```

```
Requirement already satisfied: category_encoders in /usr/local/lib/python3.10/dis
t-packages (2.6.3)
Requirement already satisfied: numpy>=1.14.0 in /usr/local/lib/python3.10/dist-pa
ckages (from category_encoders) (1.25.2)
Requirement already satisfied: scikit-learn>=0.20.0 in /usr/local/lib/python3.10/
dist-packages (from category_encoders) (1.2.2)
Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.10/dist-pac
kages (from category_encoders) (1.11.4)
Requirement already satisfied: statsmodels>=0.9.0 in /usr/local/lib/python3.10/di
st-packages (from category_encoders) (0.14.2)
Requirement already satisfied: pandas>=1.0.5 in /usr/local/lib/python3.10/dist-pa
ckages (from category_encoders) (2.0.3)
Requirement already satisfied: patsy>=0.5.1 in /usr/local/lib/python3.10/dist-pac
kages (from category_encoders) (0.5.6)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.1
0/dist-packages (from pandas>=1.0.5->category_encoders) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.10/dist-pac
kages (from pandas>=1.0.5->category_encoders) (2023.4)
Requirement already satisfied: tzdata>=2022.1 in /usr/local/lib/python3.10/dist-p
ackages (from pandas>=1.0.5->category_encoders) (2024.1)
Requirement already satisfied: six in /usr/local/lib/python3.10/dist-packages (fr
om patsy>=0.5.1->category_encoders) (1.16.0)
Requirement already satisfied: joblib>=1.1.1 in /usr/local/lib/python3.10/dist-pa
ckages (from scikit-learn>=0.20.0->category_encoders) (1.4.2)
Requirement already satisfied: threadpoolctl>=2.0.0 in /usr/local/lib/python3.10/
dist-packages (from scikit-learn>=0.20.0->category_encoders) (3.5.0)
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.10/dist-
packages (from statsmodels>=0.9.0->category_encoders) (24.0)
```

In [ ]:
```
df=pd.read_csv('modclothes_sample.csv')
```

Описание датасета

item_id: ID товара | waist: размер талии клиента | size: размера товара | quality: оценка товара | cup size: размер чашки клиента | hips: размер бедер клиента | bra size: размер бюстгальтера клиента | category: категория товара | bust: обхват груди клиента | height: рост клиента | length: отзыв о длине изделия | fit: обратная связь о соответствии размера | user_id: ID клиента | shoe size: размер обуви клиента | shoe width: ширина обуви клиента | review_text: отзыв клиента | review_summary: сводка отзыва клиента

In [ ]:
```
df.head()
```

Out[ ]:

	item_id	waist	size	quality	cup size	hips	bra size	category	bust	height	user_nam
0	234821	NaN	38	5.0	c	NaN	42.0	dresses	NaN	5ft 10in	Jasmin
1	130225	NaN	12	2.0	b	38.0	36.0	new	NaN	5ft 10in	Brit
2	175771	NaN	12	4.0	NaN	NaN	NaN	dresses	NaN	5ft 10in	ar.dive
3	154411	NaN	4	3.0	dd/e	36.0	32.0	new	34.0	5ft 2in	srwilliams021
4	175771	NaN	4	5.0	b	38.0	32.0	dresses	NaN	5ft 2in	Nandir

In [ ]:
```
df['category'].value_counts()
```

Out[ ]:
```
category
dresses    1732
new        1312
wedding      42
sale         34
Name: count, dtype: int64
```

In [ ]:
```
ce_CountEncoder1 = ce_CountEncoder(normalize=True)
data_COUNT_ENC = ce_CountEncoder1.fit_transform(df['category'])
data_COUNT_ENC
```

Out[ ]:

	category
0	0.555128
1	0.420513
2	0.555128
3	0.420513
4	0.555128
...	...
3115	0.555128
3116	0.420513
3117	0.555128
3118	0.555128
3119	0.420513

3120 rows × 1 columns

Задача 2

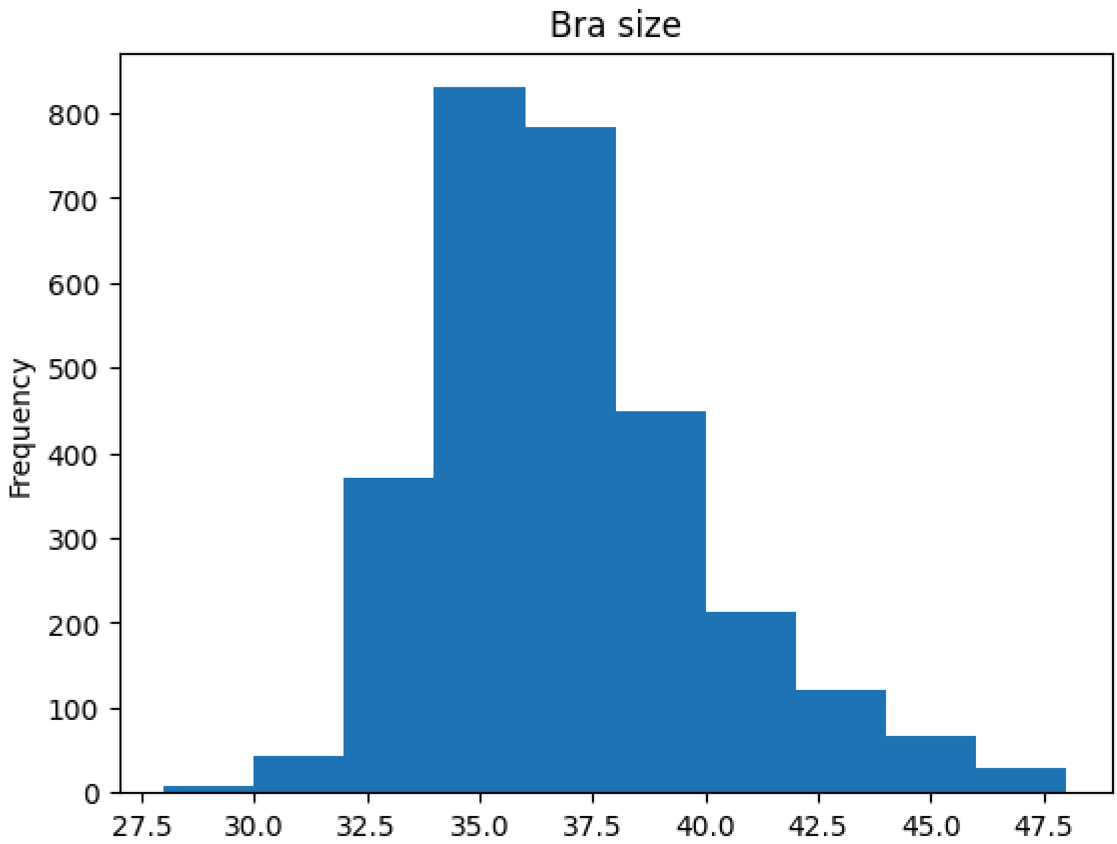
Номер задачи №2 - 21. (Для набора данных проведите масштабирование данных для одного (произвольного) числового признака с использованием масштабирования по медиане.)

In [ ]:

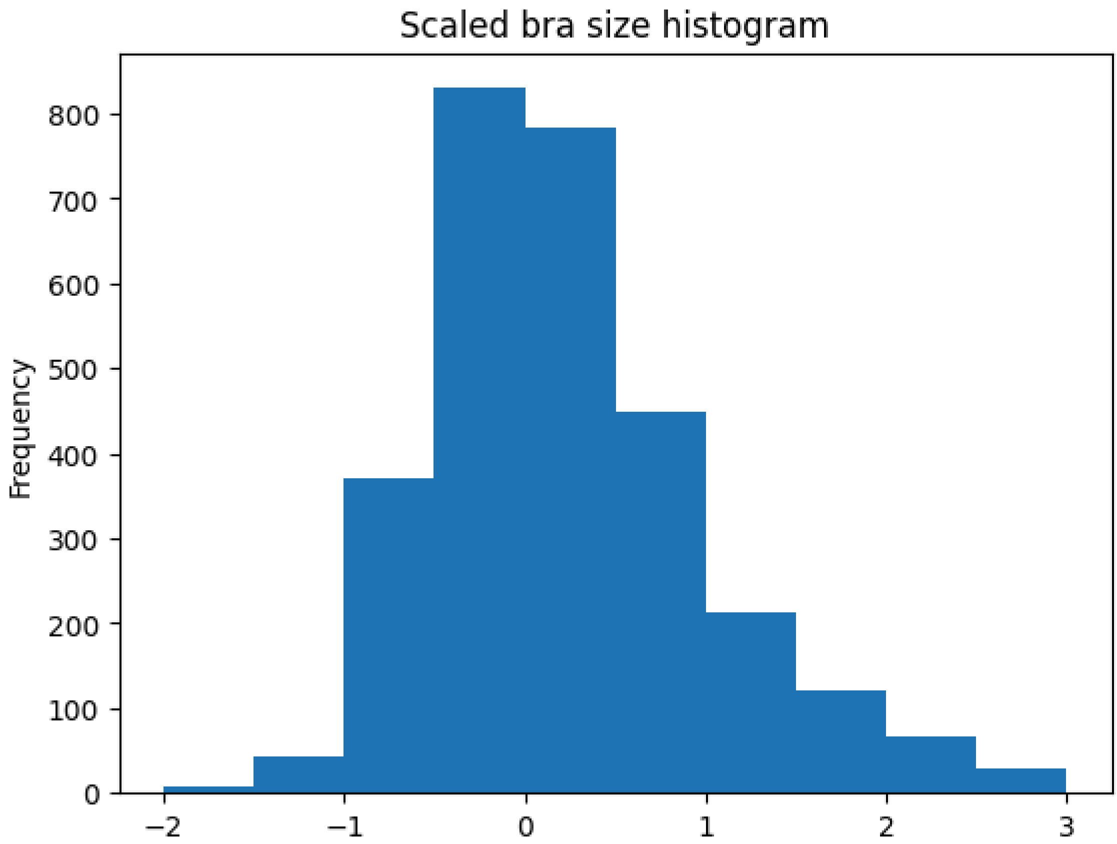
```
df['bra size'].plot.hist(title='Bra size')
```

Out[ ]:

```
<Axes: title={'center': 'Bra size'}, ylabel='Frequency'>
```



In [ ]:
```
scaler = RobustScaler()
scaled_data = pd.Series(scaler.fit_transform(df[['bra size']]).reshape (1, -1) [
scaled_data.plot.hist(title='Scaled bra size histogram');
```



In [ ]:
```
df['bra size'].median()
```

Out[ ]: 36.0

In [ ]:
```
scaled_data.median()
```

Out[ ]: 0.0

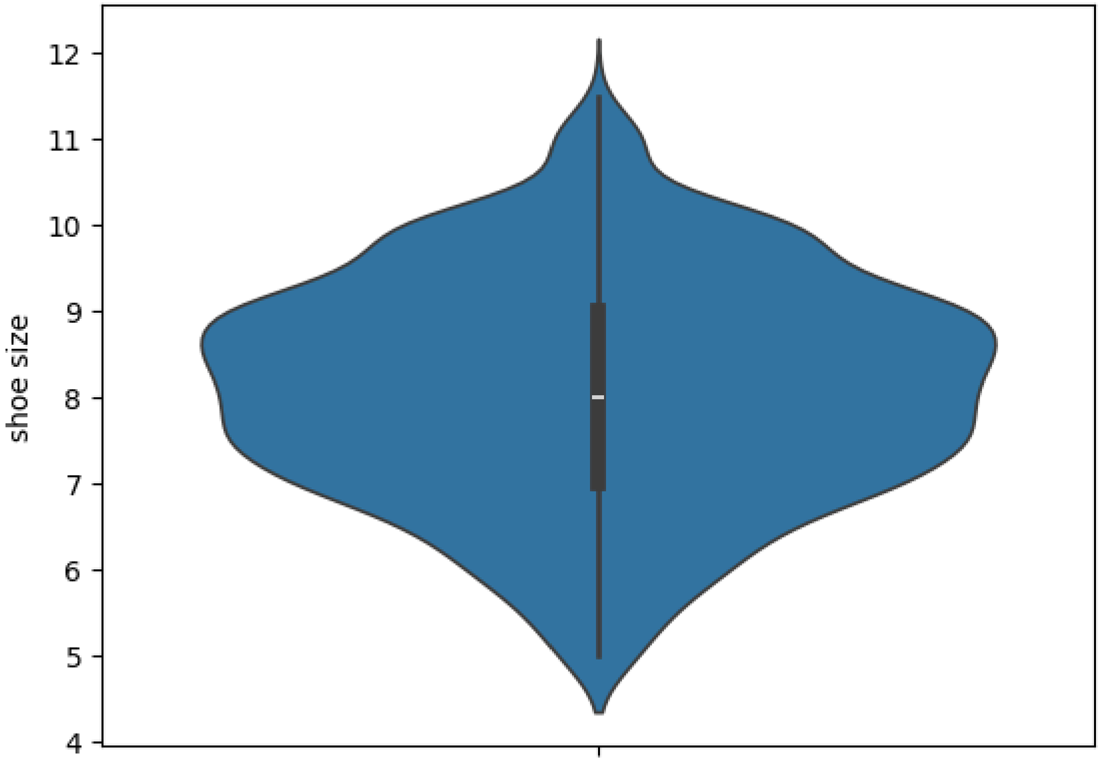
Дополнительное задание

Для произвольной колонки данных построить график "Скрипичная диаграмма (violin plot)"

In [ ]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [ ]:
```
sns.violinplot(df['shoe size'])
```

Out[ ]: <Axes: ylabel='shoe size'>



In [4]:
```
!jupyter nbconvert --to html /content/РК1_Гордеев.ipynb
```

```
[NbConvertApp] WARNING | pattern '/content/РК1_Гордеев.ipynb' matched no files
This application is used to convert notebook files (*.ipynb)
        to various other formats.

        WARNING: THE COMMANDLINE INTERFACE MAY CHANGE IN FUTURE RELEASES.

Options
=======
The options below are convenience aliases to configurable class-options,
as listed in the "Equivalent to" description-line of the aliases.
To see all configurable class-options for some <cmd>, use:
    <cmd> --help-all

--debug
    set log level to logging.DEBUG (maximize logging output)
    Equivalent to: [--Application.log_level=10]
--show-config
    Show the application's configuration (human-readable format)
    Equivalent to: [--Application.show_config=True]
--show-config-json
    Show the application's configuration (json format)
    Equivalent to: [--Application.show_config_json=True]
--generate-config
    generate default config file
    Equivalent to: [--JupyterApp.generate_config=True]
-y
    Answer yes to any questions instead of prompting.
    Equivalent to: [--JupyterApp.answer_yes=True]
--execute
    Execute the notebook prior to export.
    Equivalent to: [--ExecutePreprocessor.enabled=True]
--allow-errors
    Continue notebook execution even if one of the cells throws an error and incl
ude the error message in the cell output (the default behaviour is to abort conve
rsion). This flag is only relevant if '--execute' was specified, too.
    Equivalent to: [--ExecutePreprocessor.allow_errors=True]
--stdin
    read a single notebook file from stdin. Write the resulting notebook with def
ault basename 'notebook.*'
    Equivalent to: [--NbConvertApp.from_stdin=True]
--stdout
    Write notebook output to stdout instead of files.
    Equivalent to: [--NbConvertApp.writer_class=StdoutWriter]
--inplace
    Run nbconvert in place, overwriting the existing notebook (only
            relevant when converting to notebook format)
    Equivalent to: [--NbConvertApp.use_output_suffix=False --NbConvertApp.export_
format=notebook --FilesWriter.build_directory=]
--clear-output
    Clear output of current file and save in place,
            overwriting the existing notebook.
    Equivalent to: [--NbConvertApp.use_output_suffix=False --NbConvertApp.export_
format=notebook --FilesWriter.build_directory= --ClearOutputPreprocessor.enabled=
True]
--coalesce-streams
    Coalesce consecutive stdout and stderr outputs into one stream (within each c
ell).
    Equivalent to: [--NbConvertApp.use_output_suffix=False --NbConvertApp.export_
format=notebook --FilesWriter.build_directory= --CoalesceStreamsPreprocessor.enab
led=True]
```

```
--no-prompt
    Exclude input and output prompts from converted document.
    Equivalent to: [--TemplateExporter.exclude_input_prompt=True --TemplateExport
er.exclude_output_prompt=True]
--no-input
    Exclude input cells and output prompts from converted document.
            This mode is ideal for generating code-free reports.
    Equivalent to: [--TemplateExporter.exclude_output_prompt=True --TemplateExpor
ter.exclude_input=True --TemplateExporter.exclude_input_prompt=True]
--allow-chromium-download
    Whether to allow downloading chromium if no suitable version is found on the
system.
    Equivalent to: [--WebPDFExporter.allow_chromium_download=True]
--disable-chromium-sandbox
    Disable chromium security sandbox when converting to PDF..
    Equivalent to: [--WebPDFExporter.disable_sandbox=True]
--show-input
    Shows code input. This flag is only useful for dejavu users.
    Equivalent to: [--TemplateExporter.exclude_input=False]
--embed-images
    Embed the images as base64 dataurls in the output. This flag is only useful f
or the HTML/WebPDF/Slides exports.
    Equivalent to: [--HTMLExporter.embed_images=True]
--sanitize-html
    Whether the HTML in Markdown cells and cell outputs should be sanitized..
    Equivalent to: [--HTMLExporter.sanitize_html=True]
--log-level=<Enum>
    Set the log level by value or name.
    Choices: any of [0, 10, 20, 30, 40, 50, 'DEBUG', 'INFO', 'WARN', 'ERROR', 'CR
ITICAL']
    Default: 30
    Equivalent to: [--Application.log_level]
--config=<Unicode>
    Full path of a config file.
    Default: ''
    Equivalent to: [--JupyterApp.config_file]
--to=<Unicode>
    The export format to be used, either one of the built-in formats
            ['asciidoc', 'custom', 'html', 'latex', 'markdown', 'notebook', 'pd
f', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpdf']
            or a dotted object name that represents the import path for an
            ``Exporter`` class
    Default: ''
    Equivalent to: [--NbConvertApp.export_format]
--template=<Unicode>
    Name of the template to use
    Default: ''
    Equivalent to: [--TemplateExporter.template_name]
--template-file=<Unicode>
    Name of the template file to use
    Default: None
    Equivalent to: [--TemplateExporter.template_file]
--theme=<Unicode>
    Template specific theme(e.g. the name of a JupyterLab CSS theme distributed
    as prebuilt extension for the lab template)
    Default: 'light'
    Equivalent to: [--HTMLExporter.theme]
--sanitize_html=<Bool>
    Whether the HTML in Markdown cells and cell outputs should be sanitized.This
    should be set to True by nbviewer or similar tools.
```

```
    Default: False
    Equivalent to: [--HTMLExporter.sanitize_html]
--writer=<DottedObjectName>
    Writer class used to write the
                                    results of the conversion
    Default: 'FilesWriter'
    Equivalent to: [--NbConvertApp.writer_class]
--post=<DottedOrNone>
    PostProcessor class used to write the
                                    results of the conversion
    Default: ''
    Equivalent to: [--NbConvertApp.postprocessor_class]
--output=<Unicode>
    Overwrite base name use for output files.
                Supports pattern replacements '{notebook_name}'.
    Default: '{notebook_name}'
    Equivalent to: [--NbConvertApp.output_base]
--output-dir=<Unicode>
    Directory to write output(s) to. Defaults
                                  to output to the directory of each notebook. To
recover
                                  previous default behaviour (outputting to the c
urrent
                                  working directory) use . as the flag value.
    Default: ''
    Equivalent to: [--FilesWriter.build_directory]
--reveal-prefix=<Unicode>
    The URL prefix for reveal.js (version 3.x).
            This defaults to the reveal CDN, but can be any url pointing to a cop
y
            of reveal.js.
            For speaker notes to work, this must be a relative path to a local
            copy of reveal.js: e.g., "reveal.js".
            If a relative path is given, it must be a subdirectory of the
            current directory (from which the server is run).
            See the usage documentation
            (https://nbconvert.readthedocs.io/en/latest/usage.html#reveal-js-html
-slideshow)
            for more details.
    Default: ''
    Equivalent to: [--SlidesExporter.reveal_url_prefix]
--nbformat=<Enum>
    The nbformat version to write.
            Use this to downgrade notebooks.
    Choices: any of [1, 2, 3, 4]
    Default: 4
    Equivalent to: [--NotebookExporter.nbformat_version]

Examples
--------

    The simplest way to use nbconvert is

            > jupyter nbconvert mynotebook.ipynb --to html

            Options include ['asciidoc', 'custom', 'html', 'latex', 'markdown',
'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpd
f'].

            > jupyter nbconvert --to latex mynotebook.ipynb
```

```
        Both HTML and LaTeX support multiple output templates. LaTeX includes
        'base', 'article' and 'report'.  HTML includes 'basic', 'lab' and
        'classic'. You can specify the flavor of the format used.

        > jupyter nbconvert --to html --template lab mynotebook.ipynb

        You can also pipe the output to stdout, rather than a file

        > jupyter nbconvert mynotebook.ipynb --stdout

        PDF is generated via latex

        > jupyter nbconvert mynotebook.ipynb --to pdf

        You can get (and serve) a Reveal.js-powered slideshow

        > jupyter nbconvert myslides.ipynb --to slides --post serve

        Multiple notebooks can be given at the command line in a couple of
        different ways:

        > jupyter nbconvert notebook*.ipynb
        > jupyter nbconvert notebook1.ipynb notebook2.ipynb

        or you can specify the notebooks list in a config file, containing::

            c.NbConvertApp.notebooks = ["my_notebook.ipynb"]

        > jupyter nbconvert --config mycfg.py

To see all available configurables, use `--help-all`.
```